
Heather burning and cutting on red grouse count areas, 2017/18 to 2025/26

Satellite detection with Sentinel-2, and a first estimate of heather beetle damage

Sabeeth

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Abstract

Heather on red grouse moors is managed by rotational burning and, since the 2021 deep-peat burning regulation, increasingly by cutting. We used Sentinel-2 satellite imagery to map ground burnt or cut each season (October to 15 April) on 28 red grouse count areas (4,273 ha, 16 moors) in the North Pennines, Yorkshire Dales and North York Moors, for the nine seasons from 2017/18 to 2025/26. The method compares a summer image taken before each season with an image taken after it closes, using the Normalised Burn Ratio corrected for seasonal change. This second version follows an expert review of the first: it uses definitive count area boundaries, an after-season window that starts on 16 April when the season closes, a minimum patch of 0.1 ha, and one class for burning and cutting together. On average 171 ha (4.0% of the count area) was burnt or cut each season. Management is sporadic at count-area level, heavy in one season and light or absent the next. The North York Moors are managed more intensively than the North Pennines (6.4% against 3.7% a year), the same ordering as published national estimates. Total management did not fall after the 2021 regulation, consistent with cutting replacing burning on blanket bog. 2024/25 was the highest season (280 ha, 6.6%), during the heather beetle outbreak; this holds without the new Sentinel-2C satellite and at every threshold tested. Re-running the analysis one change at a time shows that the smaller minimum patch had the largest effect on the results. Sentinel-2 can also estimate heather beetle damage: a rule-based estimate averaged with gradient boosting, trained on labels taken automatically from the July 2025 survey, is within 0.1 of the survey on 15 of 22 count areas when each moor is predicted without its own data.

This version takes account of Phil's (PW) review of version 1 and uses Eleanor's count area polygon boundaries.

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Key findings

Main results

- **171 ha, 4.0% of the count area, was burnt or cut in an average season.** The lowest season was 2020/21 (2.5%) and the highest 2024/25 (6.6%) (Section 4.1).
- **Management is sporadic at count-area level.** Edmundbyers, Waskerley was heavily managed in four of nine seasons (37%, 12%, 41% and 40%) and barely touched in the other five. Every season is reported (Section 4.2).
- **The North York Moors are managed most intensively:** 6.4% a year against 3.7% in the North Pennines and in the Dales. This matches the ordering in Shewring et al. (2024), who report about 6.3% and 2.7% for burning alone (Section 4.3).
- **Total management held steady after the 2021 regulation:** 158 ha a season before it and 155 ha from 2021/22 to 2023/24, consistent with cutting replacing burning on blanket bog (Section 4.4).
- **2024/25 is the highest season, 280 ha (6.6%),** during the heather beetle outbreak, consistent with PW’s observation that managers burnt and cut beetle-killed heather. It remains the highest without Sentinel-2C scenes and at every threshold tested (Section 4.5).
- **The reserves behave as expected:** Moor House National Nature Reserve 0.08% a year and Geltsdale RSPB reserve 1.0%.
- **The patterns are robust.** A threshold 0.02 higher or lower moves totals by 20 to 25% but leaves the order of seasons and moors, the regional contrast and the steady post-2021 total in place (Section 4.6).
- **Sentinel-2 can estimate heather beetle damage.** The combined estimate is within 0.1 of the July 2025 survey on 15 of 22 count areas and within 0.2 on 19 (Section 6).

New in version 2	
Change	What it showed
Definitive count area boundaries (Eleanor), plus Danby, Glaisdale	Count areas are larger (4,273 ha against 2,537 ha), but the share managed is unchanged at 3.3% under the version 1 method. The results are not sensitive to the boundaries.
After-season window from 16 April (PW)	Makes sure early-April work is counted. The total barely moves (3.3% to 3.2%), because the after-season image is a median of many scenes, most taken after the work. Version 1’s “late” detections were mostly heather beetle browning in summers 2024 and 2025, not early-April burns (Section 5.2).
Minimum patch 0.1 ha instead of 0.5 ha (PW)	The largest effect: 3.3% to 4.2% a year, and count-area seasons with nothing detected fall from 35% to 25%. Many fires and cuts are small, as PW said.
One class, burnt or cut; every season reported (PW)	The sporadic pattern is real management, and the total holds steady after 2021 as cutting replaces burning.
2024/25 included and tested	Highest season under every combination of settings, without Sentinel-2C, and at every threshold.
Heather beetle damage	New: a satellite estimate tested against the July 2025 survey (Section 6).

1 Introduction

1.1 Heather management on grouse moors

Heather on moorland managed for red grouse is burnt or cut in small patches on a rotation to keep a mosaic of young heather for food and older heather for cover. Burning is contentious for its effects on peat, hydrology and habitat condition. Since 1 May 2021, burning on deep peat (more than 40 cm) in protected sites in England has needed a licence ([The Heather and Grass etc. Burning \(England\) Regulations, 2021](#)), and many estates on blanket bog have switched to cutting. Burning and cutting are both carried out in the same season, from October to 15 April.

For a study of red grouse, the useful quantity is therefore **total heather management**, burning plus cutting: how much of each count area was managed in each season. Grouse are counted on these areas, so the results can be related directly to the counts.

1.2 Measuring management from space

Burning and cutting both remove the heather canopy. This lowers near-infrared reflectance and raises short-wave infrared reflectance, which the Normalised Burn Ratio (NBR) captures ([Key and Benson, 2006](#)). Sentinel-2 provides free imagery at 10 m every few days ([Drusch et al., 2012](#)), and Google Earth Engine makes it practical to process nine years of it over many sites ([Gorelick et al., 2017](#)). [Shewring et al. \(2024\)](#) used Sentinel-2 with a deep neural network to map burning across Great Britain from 2017/18 to 2021/22. They found about 6.3% of the North York Moors and 2.7% of the North Pennines burnt each year, and a 73% fall in burning in England in 2021/22 after the regulation. Their study measured burning only.

1.3 Version 1 and the expert review

Version 1 of this analysis used the same Sentinel-2 approach on 27 count areas reconstructed from transect lines. PW, who knows these moors in the field, reviewed it and made eight main points:

1. The burning season runs to 15 April, and early April is often the busiest period, so the after-season image should start on 16 April.
2. Many fires and cuts are much smaller than the 0.5 ha minimum, sometimes 10 m by 20 m.
3. Burning and cutting are both carried out from October to 15 April, so they cannot be separated by timing.
4. Every season is usable: management is sporadic, intensive in one year and absent the next.
5. After 2021, estates on blanket bog cut instead of burning, so total management need not fall.
6. 2024/25 is high because managers burnt and cut beetle-killed heather after a large heather beetle outbreak.
7. More cutting was expected on Crossgill and Wemmergill than on Geltsdale; these sites practise small cuts.
8. Definitive count area boundaries are available, and the map in version 1 did not show the proportion managed clearly.

Version 2 takes up every point. Section 5 shows the effect of each change and responds point by point.

1.4 Aims

1. Measure total heather management (burning plus cutting) on each count area in each season from 2017/18 to 2025/26, using the definitive boundaries.
2. Apply the changes from the review and measure the effect of each one.
3. Test how robust the main findings are to the detection threshold, the satellite supply and the boundaries.
4. Assess whether Sentinel-2 can measure heather beetle damage, using the July 2025 survey as the reference.

1.5 What was done

1. **Count areas.** Eleanor's definitive polygons were named from the 2026 transect file and checked. One count area is new (Danby, Glaisdale), and Eggleston's single block was split into its seven count areas (Section 2.1).
2. **Detection.** The detection was re-run for nine seasons with the 16 April window, a 0.1 ha minimum patch and one "burnt or cut" class (Section 3.2).
3. **Robustness.** The analysis was repeated at four thresholds, without Sentinel-2C for 2024/25, and with Eggleston as one block (Section 4.6).
4. **Effect of each change.** Version 1's boundaries were rebuilt, its results were reproduced, and the analysis was re-run with one change at a time (Section 5).
5. **Imagery.** Season-by-season image sheets were made for six count areas to check what the detections look like (Section 4.2).
6. **Heather beetle damage.** A rule-based estimate and a gradient boosting model were built from the imagery, compared with the survey on moors they had not seen, and checked for data leakage (Section 6).

2 Study area and data

2.1 Count areas

The count area boundaries are the definitive polygons digitised by Eleanor (GWCT) in October 2026. Long-term counts are 1 km² blocks walked as six parallel transects 150 m apart. Other counts are two parallel transects, typically 3 km long and 500 m apart, and their count area is the transects plus a 150 m buffer.

The polygon file carried no names, so each polygon was named from the count transects it contains (the 2026 transect file). This gives **28 count areas, 4,273 ha, on 16 moors** (Table 1). Count areas range from 54 to 290 ha (median 138 ha). One count area is new since version 1: **Danby, Glaisdale**.

Eggleston was supplied as a single 1,928 ha block covering all seven of its count areas. We split it with the same rule (two transects plus 150 m). That rule reproduces Eleanor's other 13 transect polygons almost exactly (overlap 0.99 to 1.00). The 371 ha strip between the Eggleston count areas is measured but left out of all totals. This makes little difference: Eggleston is 5.2% a year as seven count areas and 5.3% as one block (Appendix D).

Eggleston's seven count areas make up 1,557 ha, over a third of the total, so area-weighted figures

Table 1. Count areas by region.

Region	Moors	Count areas	Area (ha)
North Pennines	7	18	3,237
North York Moors	4	5	508
Yorkshire Dales and Nidderdale	5	5	529
All	16	28	4,273

Areas are rounded to the nearest hectare.

for the North Pennines lean on Eggleston. Results are therefore also given by moor and by count area.

2.2 Satellite imagery

We used Sentinel-2 Level-2A surface reflectance (10 m for the bands used in detection) in Google Earth Engine. Sentinel-2A and 2B supplied the imagery throughout. Sentinel-2C entered service during 2024/25 and raised the number of after-season scenes from about 106 to 156 (Section 4.5). The after-season image had a median of 7.5 clear scenes per pixel. The fewest were at Grinton, Long Gill in 2024/25, with about one, so that value is less certain.

2.3 Heather beetle survey

A survey in July 2025 estimated, by eye, the proportion of each count area damaged by heather beetle. It scored 23 count areas in tenths, from 0 to 0.8. At Eggleston only Blackton and Ever Rigg were scored.

3 Methods

3.1 Overview

Figure 1 summarises the detection. The formulae are in Appendix A.

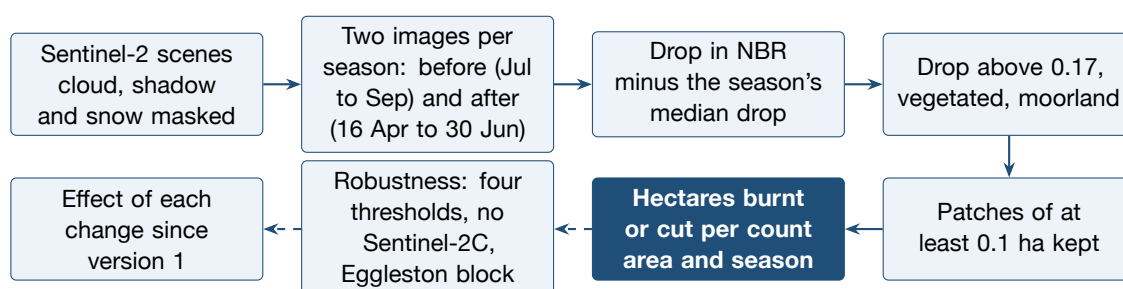


Figure 1. Detection of burnt or cut ground (solid arrows) and the tests applied to the results (dashed arrows).

3.2 Detecting burnt or cut ground

Imagery and index. Clouds, cloud shadow and snow were masked with the s2cloudless cloud probability (below 40%) and the scene classification layer. The index is the Normalised Burn

Ratio, $NBR = (NIR - SWIR2)/(NIR + SWIR2)$, from bands B8 and B12. It falls sharply when vegetation is burnt or cut.

Two images per season (Table 2), each the median of all clear scenes in its window. The drop in NBR between them (dNBR) is the signal.

Table 2. Image windows. Version 1 used an after window of 1 April to 31 May, which overlapped the last two weeks of the season.

Image	Window	Purpose
Before	1 July to 30 September, before the season	Vegetation before any work that season
After	16 April to 30 June, after the season	Ground burnt or cut during the season

Seasonal correction. Heather browns over winter whether or not it is managed. Only a few per cent of the count areas are managed in any season, so the median dNBR across all count areas measures this background change. It is subtracted each season. Afterwards the typical pixel shows no change (median between -0.004 and 0.004 in every season).

Threshold. A pixel counts as managed if its corrected dNBR exceeds **0.17**, the 95th percentile of dNBR in 2018/19 used in version 1. Re-checked on the new boundaries, the 2018/19 value is 0.176 against 0.168 , so the calibration holds. The 95th percentile varies between seasons (0.135 to 0.207) because the amount of management varies. Matching the threshold to each season would force the same total every season, so one threshold is used throughout.

Vegetation and land cover. A pixel must have been vegetated in the before image (NDVI above 0.3). Trees, crops, buildings and water are masked using ESA WorldCover (Zanaga et al., 2022). Inside the count areas WorldCover classes 98.5% of the ground as grassland and none as shrubland, so this mask removes only the 1.5% of trees. It does not separate heather from bog.

Minimum patch: 0.1 ha (ten connected 10 m pixels), the practical floor for this imagery. The smallest cuts, around 10 m by 20 m, are still below it.

One class. What distinguishes a cut from a burn (straight edges, tracks a few metres wide) is not visible at 10 m, and both happen in the same season. Results are therefore “burnt or cut”.

3.3 Robustness tests

- **Threshold:** the full analysis was repeated at 0.13, 0.15 and 0.19.
- **Sentinel-2C:** 2024/25 was re-run without Sentinel-2C scenes.
- **Eggleston boundaries:** seven count areas against Eleanor’s single block.

3.4 Separating the effect of each change since version 1

Version 2 changed the boundaries, the after window and the minimum patch at the same time. To separate them, version 1’s count areas were rebuilt exactly as version 1 built them: the closed ring where a line was drawn as one, otherwise the convex hull of the count’s lines. Their areas match version 1’s to within 0.5%. The detection was then run with each combination of boundaries (version 1 or 2), after window (1 April to 31 May, or 16 April to 30 June) and minimum patch (0.5 or 0.1 ha). The comparison uses the 27 count areas common to both versions and the eight seasons version 1 covered (2017/18 to 2024/25).

3.5 Heather beetle damage

Beetle-killed heather turns red-brown in the summer it is attacked, then grey as the dead leaves drop. The estimate is built in two steps.

Step 1: a rule-based estimate. Two changes capture the damage:

- **Browning:** the drop in NBR from late summer 2024 to late summer 2025.
- **Greying:** the drop in colour saturation of the visible bands from early to late summer 2025. Flatter colour is greyer.

Summer 2025 was dry and the whole area browned, so the median change across all count areas is subtracted. A pixel is **flagged** if it browned by more than 0.10 or greyed by more than 0.05 beyond that median. Ground burnt or cut in 2024/25 or 2025/26, plus a 20 m margin, is left out. The estimate is the **share of each count area flagged**; nothing is fitted to the survey.

Step 2: gradient boosting with labels taken from the survey. About 280 pixels were sampled in each count area. Each pixel is described by 45 changes: all ten Sentinel-2 bands (including the three red-edge bands) and five indices, each over three time comparisons. Pixels from count areas the survey scored 0 were labelled healthy, and pixels from count areas scored 0.5 or more were labelled damaged. No labels were drawn by hand. A gradient boosting classifier (Friedman, 2001) learns from these labels, and each count area's mean probability is converted to a proportion with a straight line. The **final estimate is the average of the two steps**.

Testing. Every estimate is made by a model that has not seen the moor being predicted (leave-one-moor-out; Roberts et al., 2017). Agreement with the survey is summarised as the typical difference (mean absolute difference), the number of count areas within 0.1 and within 0.2, and the rank agreement (Spearman's ρ). Grinton, Long Gill is left out because it had too few clear images, leaving 22 count areas.

4 Results: heather management

4.1 Season by season

Table 3. Ground burnt or cut by season.

Season	Burnt or cut (ha)	% of count area	Clear images per pixel ^a
2017/18	169	4.0	11.2
2018/19	201	4.7	5.5
2019/20	156	3.7	8.2
2020/21	106	2.5	8.6
2021/22	199	4.7	4.9
2022/23	153	3.6	6.7
2023/24	114	2.7	4.8
2024/25	280	6.6	9.1
2025/26	158	3.7	7.7
Mean	171	4.0	7.5

^a Median over count areas of the clear after-season scenes per pixel; the mean row gives the median over all count-area seasons.

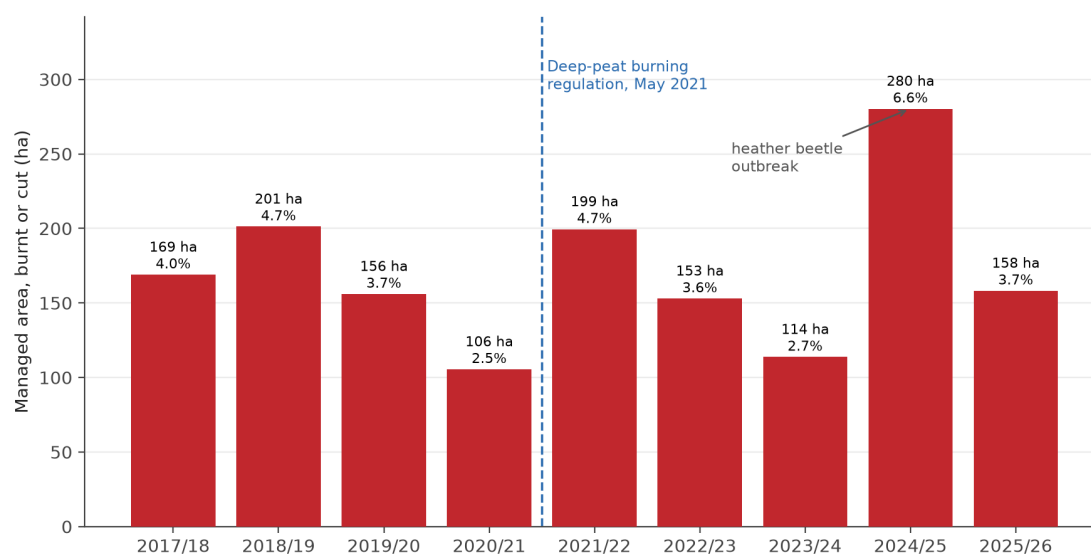


Figure 2. Ground burnt or cut within the 28 count areas, by season. The dashed line marks the deep-peat burning regulation of May 2021.

On average **171 ha, 4.0% of the count area, was burnt or cut each season** (Figure 2, Table 3). Seasons range from 106 ha (2.5%) in 2020/21 to 280 ha (6.6%) in 2024/25.

4.2 Count areas and moors

Figure 3 shows the **sporadic pattern** PW described. Edmundbyers, Waskerley was heavily managed in four seasons (37%, 12%, 41% and 40%) and barely touched in the other five. Snilesworth, Lodge shows the same pattern. Most other working moors manage a few per cent of each count area in most seasons.

The imagery shows what a heavy season looks like (Figure 4). At Waskerley the managed ground in each heavy season is many small patches spread across the count area: the mosaic of rotational burning and cutting, not a single event.

One single event stands out: **Bollihope, Pikestone Fell, 41% in 2022/23**, against 0 to 8% in every other season (version 1 also found it). Unlike Waskerley's mosaic, it is one solid block of about 40 ha, far larger than a normal rotational burn of 2 to 5 ha (Figure 5). It may have been a wildfire and is worth checking with the estate.

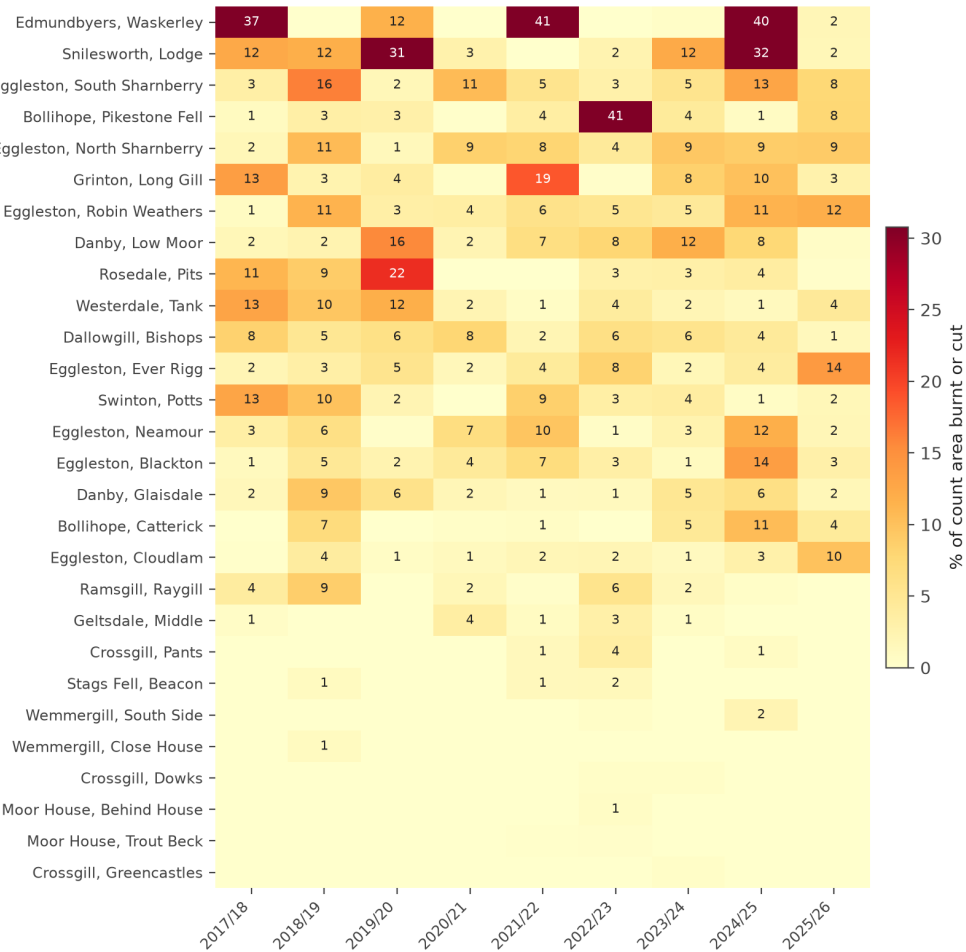


Figure 3. Percentage of each count area burnt or cut, by season. Values of 0.5% or more are printed. Count areas are ordered by their mean.

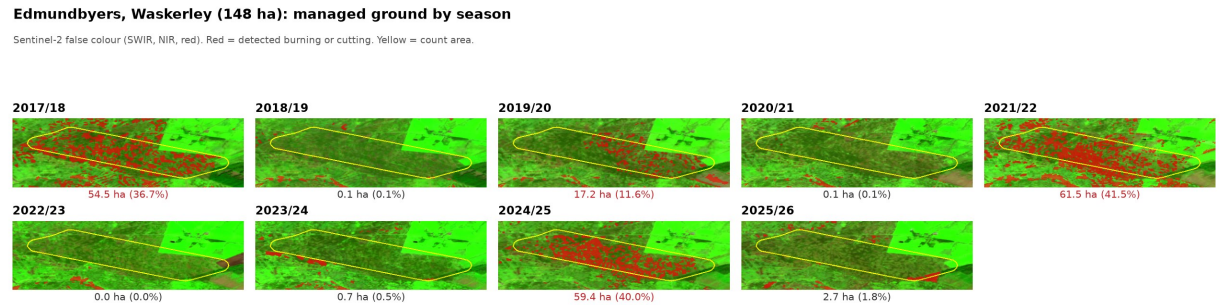


Figure 4. Edmundbyers, Waskerley, season by season. Sentinel-2 false colour after each season; red is ground detected as burnt or cut; yellow is the count area.

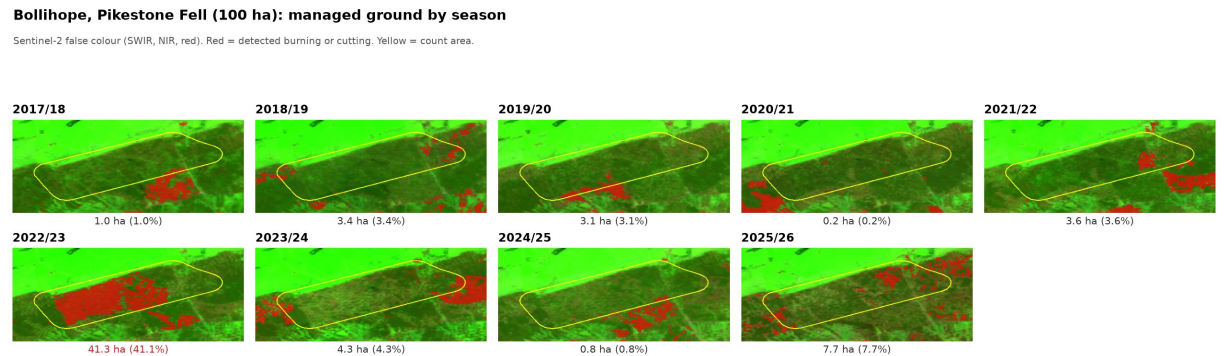


Figure 5. Bollihope, Pikestone Fell, season by season. The 2022/23 detection is a single block of about 40 ha.

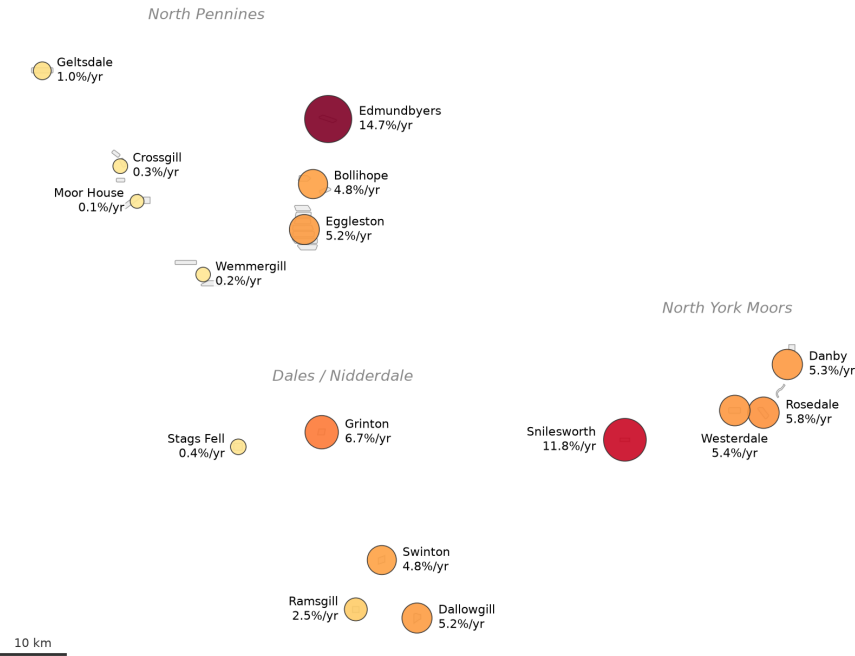


Figure 6. Mean annual percentage of count area burnt or cut, by moor. Circle size and colour show the percentage.

Table 4. Mean annual percentage of count area burnt or cut, by moor.

Moor	Region	Count areas	Area (ha)	Mean % a year
Edmundbyers	North Pennines	1	148	14.7
Snilesworth	North York Moors	1	73	11.8
Grinton	Dales and Nidderdale	1	91	6.7
Rosedale	North York Moors	1	139	5.8
Westerdale	North York Moors	1	153	5.4
Danby	North York Moors	2	142	5.3
Dallowgill	Dales and Nidderdale	1	112	5.2
Eggleston	North Pennines	7	1,557	5.2
Bollihope	North Pennines	2	237	4.8
Swinton	Dales and Nidderdale	1	97	4.8
Ramsgill	Dales and Nidderdale	1	92	2.5
Geltsdale	North Pennines	1	263	0.98
Stags Fell	Dales and Nidderdale	1	137	0.45
Crossgill	North Pennines	3	225	0.26
Wemmergill	North Pennines	2	496	0.23
Moor House	North Pennines	2	310	0.08

Table 5. Mean annual percentage burnt or cut by region, against [Shewring et al. \(2024\)](#).

Region	This study (burnt or cut)	Shewring et al. (2024) (burnt only)
North York Moors	6.4% a year	about 6.3% a year
North Pennines	3.7% a year	about 2.7% a year
Yorkshire Dales and Nidderdale	3.7% a year	not compared

By moor (Figure 6, Table 4), Edmundbyers (14.7% a year) and Snilesworth (11.8%) are the most intensively managed. Five moors are at or below 1%: Geltsdale, Stags Fell, Crossgill, Wemmergill and Moor House. Grinton's 6.7% rests partly on two seasons with very few clear images (2021/22 and 2024/25).

4.3 Regional pattern

The ordering matches Shewring et al. (2024), and the North York Moors figure is almost identical (Table 5). The North Pennines figure is higher than theirs. Our figure includes cutting and theirs does not, and with cutting replacing burning on blanket bog since 2021 a higher figure is expected. The two studies also cover different areas: count areas here, whole regions there.

4.4 Before and after the 2021 regulation

Table 6. Mean ground burnt or cut per season, by period.

Period	Burnt or cut per season
2017/18 to 2020/21 (before)	158 ha (3.7%)
2021/22 to 2023/24 (after)	155 ha (3.6%)
2024/25 (beetle outbreak)	280 ha (6.6%)
2025/26	158 ha (3.7%)

Total management held steady after the regulation (Table 6). This is consistent with PW's field observation that estates on blanket bog switched from burning to cutting, which is not limited by weather. It is also compatible with the 73% fall in burning in Shewring et al. (2024), which does not count cutting. The satellite measures burning and cutting together, so it shows the total holding steady; the field observations show the switch within it.

4.5 The 2024/25 peak

2024/25 is the highest season in the record. Without Sentinel-2C scenes it is 252 ha, 10% lower but still well above the next highest season (201 ha), and the ranking of count areas barely changes (rank correlation 0.97). It is the highest season at every threshold, and under every combination of boundaries, window and patch size tested in Section 5.

The peak coincides with the heather beetle outbreak and fits PW's observation that managers burnt and cut beetle-killed heather to promote recovery. Count areas with more damage in the July 2025 survey were more heavily managed in 2024/25 (rank correlation 0.55). They were also the more heavily managed ones before the outbreak (0.70 in 2023/24), and relative to each count area's usual level the 2024/25 increase is spread across count areas. Damage records from 2024, the summer before managers responded, would allow the response to be traced count area by count area.

Beetle-related management may also be under-counted. Burning or cutting heather that is already dead changes the satellite signal less than burning live heather. This may be why the count areas with the heaviest damage show less management than usual in 2025/26 (rank correlation -0.40 relative to their usual level).

4.6 Robustness

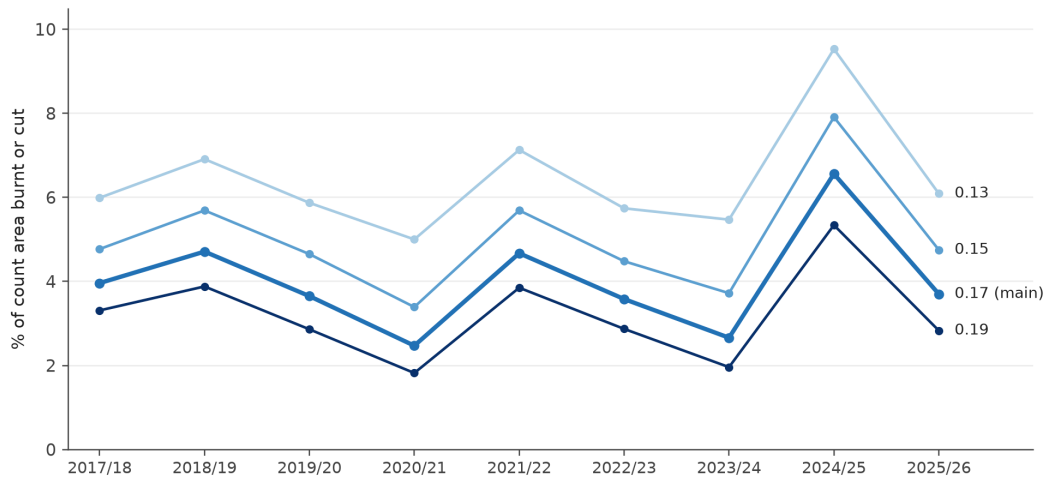


Figure 7. Percentage of count area burnt or cut by season at four thresholds. Levels shift; the pattern does not.

Between thresholds of 0.15 and 0.19 the totals change by about a fifth, but every pattern in this report holds (Figure 7). The order of seasons and moors agrees with the main run (rank correlations 0.93 to 1.00). The North York Moors stay 1.6 to 1.8 times the North Pennines, and the after/before 2021 ratio stays between 0.97 and 1.03. At 0.13 the reserves start to register (Geltsdale 4.5% a year), so 0.13 is too permissive. The full table is in Appendix B.

Table 7. Checks on the results.

Check	Expected	Result
Moor House (National Nature Reserve)	Near zero	0.08% a year
Geltsdale (RSPB reserve)	Low	1.0% a year
North York Moors against North Pennines	NYM higher (Shewring et al., 2024)	6.4% against 3.7%
Crossgill and Wemmergill	More than Geltsdale (PW)	0.26% and 0.23%: their cuts are mostly smaller than 0.1 ha, as PW anticipated
2024/25 and Sentinel-2C	Similar without it	252 against 280 ha, still the highest
Eggleston as one block	Similar	5.3% against 5.2%

5 What the review changed

5.1 Effect of each change

Figure 8 and Table 8 show the analysis re-run one change at a time on the 27 count areas the two versions share.

- **The rebuild reproduces version 1:** 3.3% a year, 41% of count-area seasons with nothing detected, and 213 ha against 211 ha on the seven unchanged count areas. The comparison starts from a faithful copy.

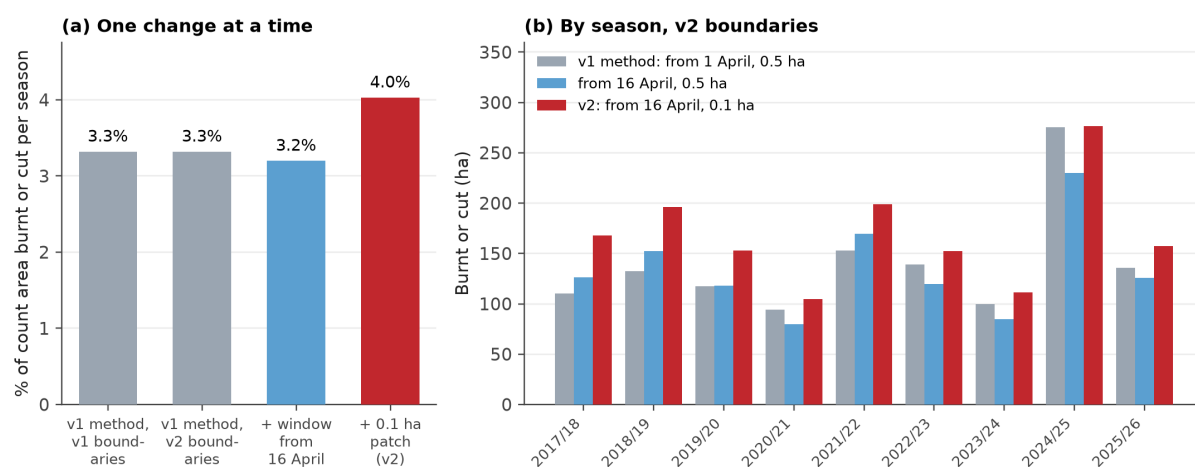


Figure 8. (a) Mean percentage burnt or cut per season on the 27 shared count areas, 2017/18 to 2024/25, as each change is added. (b) Hectares by season on the same count areas with version 2 boundaries: the version 1 method, the new window alone (16 April to 30 June), and version 2.

Table 8. The analysis with one change at a time: 27 count areas, 2017/18 to 2024/25.

Run	After window	Min. patch (ha)	% a year	Seasons with none (%)	Seven areas (ha) ^a	2024/25 (ha)
<i>Version 1 boundaries (2,546 ha)</i>						
Version 1 as published (main class)	1 Apr to 31 May	0.5	3.3	41	211	175
Version 1 method, rebuilt here	1 Apr to 31 May	0.5	3.3	41	213	175
Version 2 method	16 Apr to 30 Jun	0.1	4.2	29	284	178
<i>Version 2 boundaries (4,218 ha)</i>						
Version 1 method	1 Apr to 31 May	0.5	3.3	35	219	275
start on 16 April only	16 Apr to 31 May	0.5	3.2	33	230	252
start on 16 April, end on 30 June	16 Apr to 30 Jun	0.5	3.2	34	215	230
0.1 ha patch only	1 Apr to 31 May	0.1	4.2	25	302	319
Version 2	16 Apr to 30 Jun	0.1	4.0	23	291	277

^a Total over the eight seasons on Dallowgill, Ramsgill, Swinton, Grinton, Snilesworth, Danby Low Moor and Moor House Behind House, whose boundaries barely changed.

- **The boundaries change the hectares, not the percentages.** On Eleanor’s boundaries the count areas cover 4,218 ha instead of 2,546 ha, but the share managed stays at 3.3%. Fewer seasons show nothing (35%), because larger areas are less likely to be untouched.
- **Starting the after window on 16 April changes little on its own:** 3.3% to 3.2% a year. On the seven unchanged count areas it adds 5% (219 to 230 ha), consistent with some early-April work now being counted in full. Most early-April work was already counted, because the after image is the median of many scenes, most of them taken after the work.
- **Ending the window on 30 June rather than 31 May moves ground between seasons** without changing the total (3.2%; Figure 8b). The extra June scenes change which scars stand out in some seasons.
- **The 0.1 ha minimum patch has the largest effect.** On its own it raises the figure from 3.3% to 4.2% a year and cuts the seasons with nothing detected from 35% to 25%. Many fires and cuts are small, as PW said.
- **With all the changes** the figure is 4.0% a year, and 23% of count-area seasons show nothing.

5.2 Version 1’s late detections

Version 1 also reported a second, “late” class: ground that changed only between the after window and late summer. PW suggested these were the season’s early-April burns. On the seven unchanged count areas, version 1 found 211 ha in its main class and 65 ha in the late class, close to version 2’s 291 ha. The runs above show that starting the window on 16 April adds little on its own, so the late class was not mainly early-April work.

The timing points elsewhere. Over half of the late class (91 of 165 ha) falls in 2023/24 and 2024/25, whose late-summer images are from summers 2024 and 2025, during the beetle outbreak. In those two seasons 88 of its 91 ha lie on count areas the July 2025 survey scored 0.3 or more, and it follows the survey’s damage (rank correlation 0.51 and 0.50, against –0.06 in 2022/23). **Most of version 1’s late detections were probably heather beetle browning.** Version 2 does not count them as management; the beetle damage is now measured separately (Section 6).

5.3 Review points and responses

Table 9. PW's review of version 1, what version 2 does, and what it showed.

PW's review	Version 2	What it showed
The season ends on 15 April; early April is often the busiest	After window from 16 April	Early-April work is fully counted; totals move little (Section 5.1)
Many fires and cuts are small, sometimes 10 m by 20 m	Minimum patch 0.1 ha	The largest single effect: 3.3% to 4.2% a year
Burning and cutting both happen October to 15 April	One class, burnt or cut	Timing cannot separate them; aerial imagery can
Every season is usable; management is sporadic	Every season reported	The heavy-then-light pattern is confirmed in the imagery (Figure 4)
After 2021, blanket bog estates cut instead of burning	Total management reported	Total holds steady after 2021 (Section 4.4)
2024/25 reflects management of beetle-killed heather	Included and tested	Highest season under every test (Section 4.5)
More cutting expected at Crossgill and Wemmergill	0.1 ha patch; image sheets	Still 0.2 to 0.3%: the cuts are mostly below 0.1 ha
The late (blue) scars are early-April burns	Window from 16 April	Late detections were mostly beetle browning (Section 5.2)
Definitive boundaries available	Eleanor's polygons	Percentages unchanged; hectares larger
Map did not show the proportion	Circles sized by percentage, plus a table	Figure 6 and Figure 3

6 Heather beetle damage

6.1 Agreement with the survey

Table 10. Agreement between the satellite estimate and the survey, 22 count areas, each moor predicted without its own data.

	Region average ^a	Rule-based estimate	Rule + gradient boosting
Typical difference from the survey	0.17	0.12	0.10
Within 0.1 of the survey		12 of 22	15 of 22
Within 0.2 of the survey		19 of 22	19 of 22
Rank agreement (Spearman's ρ)		0.81	0.88

^a Each count area predicted as the average damage of the other moors in its region, with no satellite data.

The combined estimate is **within 0.1 of the survey on 15 of 22 count areas and within 0.2 on 19**, with a typical difference of 0.10 (Figure 9, Table 10). It ranks the count areas almost as the survey does (0.88). Values for every count area are in Appendix C.4.

6.2 What the satellite sees

- **The rule captures the full range of damage.** Close matches include Danby, Low Moor (0.51 against 0.5), Eggleston, Ever Rigg (0.31 against 0.3) and Ramsgill, Raygill (0.22 against 0.2). Greying matters: Dallowgill, Bishops (survey 0.6) barely browned but greyed strongly.
- **Gradient boosting learns what is not damage.** At Moor House, Trout Beck the rule reads 0.55 against a survey score of 0, because the whole bog changed colour. Gradient boosting learned from the undamaged count areas that this greying is not damage and reads 0.05; the average

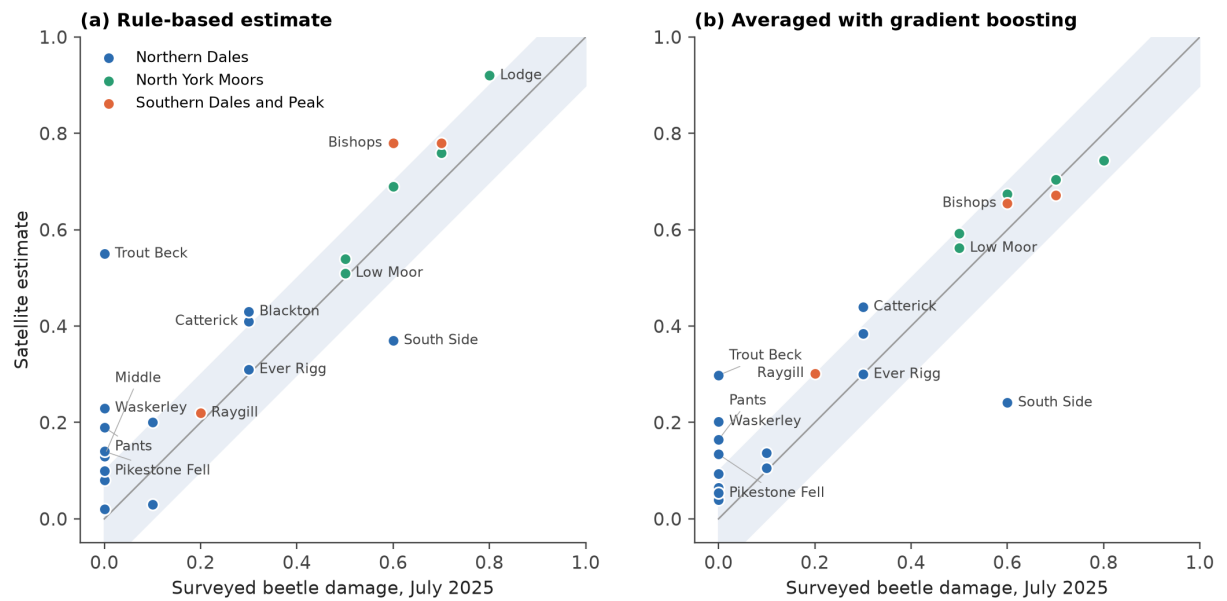


Figure 9. Satellite estimates of heather beetle damage against the July 2025 survey: (a) the rule-based estimate; (b) the rule-based estimate averaged with gradient boosting. Each moor is predicted without its own data. The shaded band is within 0.1 of the survey. Count areas more than 0.1 from the survey, or discussed in the text, are labelled.

- is 0.30.
- **The red-edge bands carry much of the signal.** After the change in NBR between early and late summer 2025, the measures gradient boosting relies on most are the red-edge bands B5 and B6, which respond to the loss of chlorophyll. Four of its ten most-used measures are red-edge (Figure 10).

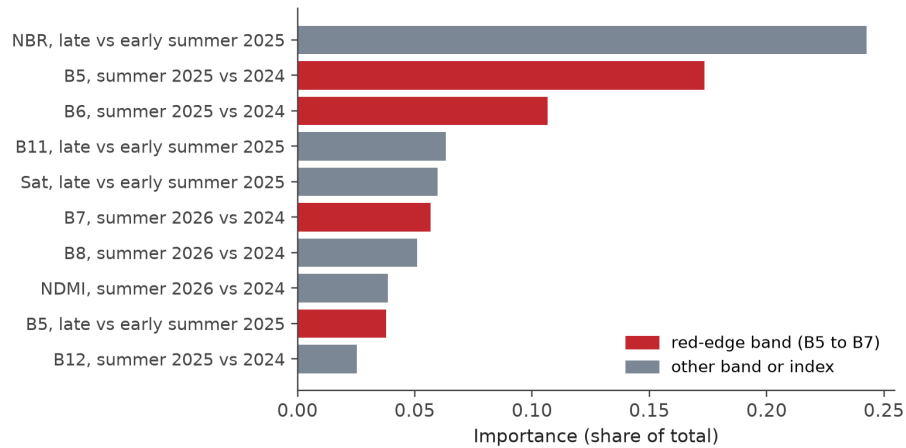


Figure 10. The ten measures gradient boosting relies on most. Each is the change in a band or index between two periods.

- **The damage is not clearly visible a year later.** Summer 2026 against summer 2024 shows no clear relationship with the survey, either because the heather recovered or because damaged heather was burnt or cut in 2025/26 and is masked out.

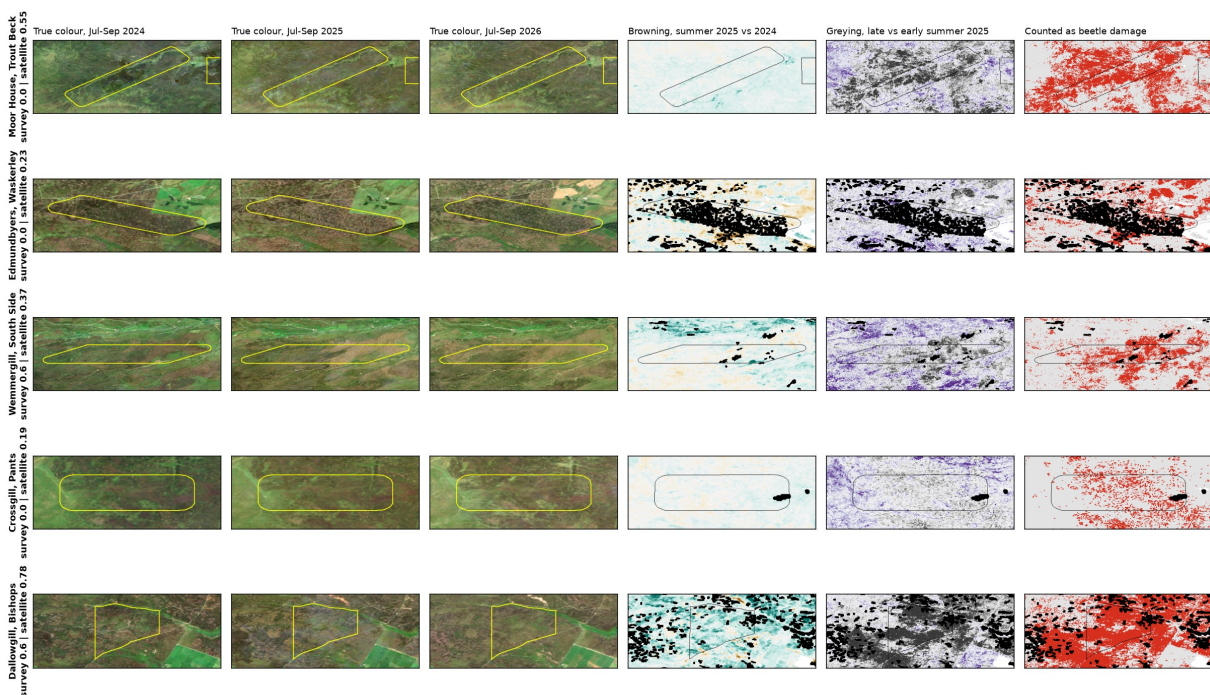
6.3 How far the estimate can be trusted

The model never sees the moor it predicts, the line converting probability to a proportion is fitted on held-out predictions, and all measures are image changes, not survey scores. Two further tests look for optimism that the design cannot rule out (Appendix C.3):

- **Choosing the model after the fact.** Six models were compared (Appendix C.2). When the model is chosen inside each fold from the training moors only, the combined estimate scores 0.105 against 0.098, so the choice adds very little.
- **Similar neighbouring moors.** When a whole region is hidden from training, gradient boosting alone does markedly worse (0.22 against 0.12 on those 8 count areas), but the combined estimate holds up (0.07, close to the rule's 0.08) because the rule needs no training. This is why the combined estimate, not the model alone, is reported.

6.4 Imagery where satellite and survey differ most

Heather beetle: the 5 count areas where satellite and survey differ most



Maps: brown = browned beyond the moor-wide change; grey = greyed beyond it; teal / purple = the opposite; red = counted as beetle damage; light grey = moorland not counted; black = burnt or cut in 2024/25 or 2025/26 plus a 20 m margin (left out); white = not moorland. Survey: July 2025 proportion damaged. Satellite: share of usable count area flagged red. Nothing is fitted to the survey.

Figure 11. The five count areas where the rule-based estimate and the survey differ most. Left to right: true colour 2024, 2025 and 2026; browning; greying; pixels counted as damage (red). Black is burnt or cut ground plus a 20 m margin.

At 10 m the true-colour images show the damage only faintly; the change maps say more (Figure 11). At these count areas the imagery mostly supports the survey:

- **Moor House, Trout Beck** (survey 0, rule 0.55, combined 0.30): the greying covers the whole bog, inside and outside the count area. It does not look like beetle damage, and gradient boosting discounts it.
- **Edmundbyers, Waskerley** (survey 0, rule 0.23): heavily burnt in 2024/25; the remaining flags

- sit around the burns.
- **Wemmergill, South Side** (survey 0.6, rule 0.37, combined 0.24): some flagged patches inside the count area, but fewer than the survey suggests. Field photos would settle it.

7 Discussion

7.1 Total management is the right measure

Since 2021 the question for grouse moors is no longer only how much is burnt, but how much is managed. Burning and cutting look alike at 10 m and happen in the same season, so the satellite measures their sum. That is a limitation for a study of burning, but it is the quantity that matters for grouse and for habitat structure. Read this way, the steady total after 2021 and the 2024/25 peak both fit what managers report.

7.2 Agreement with published work

The regional ordering and the North York Moors level match [Shewring et al. \(2024\)](#), derived independently with a different method. The higher North Pennines figure and the absence of a post-2021 fall are both expected when cutting is included. A direct test would compare the two methods on the same ground for 2017/18 to 2021/22, burning only, on moors where cutting is rare.

7.3 Using the results with grouse counts

Because management is sporadic, a count area’s figure in a single season can be large or zero. Every season is valid, and season-specific values can be related to the following summer’s count. For a measure of management intensity, the mean over several seasons is more stable. The results file holds both: hectares and percentage for each count area in each season.

7.4 Limits and how to address them

Table 11. Limits of the analysis and how to address them.

Limit	Effect	How to address it
Cuts below 0.1 ha	Small cuts are missed; Crossgill and Wemmergill read low	Aerial imagery (APGB, 25 or 50 cm) or PlanetScope (3 m)
No accuracy figure yet	The threshold is calibrated and stable, but not validated	Ground truth at 15 to 20 known burn and cut sites
Burning and cutting together	Their shares cannot be reported separately	Aerial imagery: cuts have straight edges and tracks
Management of dead heather	Likely under-counted in beetle years	Damage records from 2024; ground truth in damaged areas
Uneven image supply	A few count-area seasons rest on very few clear images (Grinton 2024/25)	Flagged in the results; PlanetScope would fill gaps
Beetle model: one year, 22 count areas	Differences of 0.01 to 0.02 between methods are within chance	A second survey year; a few hundred hand-checked patches

8 Conclusions and recommendations

Sentinel-2 records total heather management on red grouse count areas consistently from season to season. About 4% of the count area is burnt or cut in an average season, with sporadic heavy seasons on individual count areas, a higher level on the North York Moors, a steady total after the 2021 regulation and a peak in 2024/25 during the heather beetle outbreak. These patterns hold under every test applied. PW's review changed how the results are read and improved the method; of the individual changes, the smaller minimum patch mattered most. Sentinel-2 can also estimate heather beetle damage to within about 0.1 of a field survey.

Immediate

1. **Ground-truth the method.** Identify 15 to 20 sites where burning or cutting is known (estate records or field visits), digitise them and run the detection against them. This gives an accuracy figure and is the most important next step.
2. **Beetle damage evidence:** field photos from the July 2025 survey (ideally with locations), any damage scores from 2024, and a note of how the scores were estimated.
3. **Check two single results** with the estates: the 40 ha block at Pikestone Fell in 2022/23 (possibly a wildfire), and Geltsdale's detections (1.0% a year).

Next

4. **Aerial imagery** (APGB, 25 or 50 cm, colour infrared). At this resolution cuts and burns can be told apart by their shape, and small cuts become visible. Check whether GWCT already has access.
5. **Develop the beetle damage model.** A second survey year would test it on new data, and a few hundred hand-checked patches (beetle-brown, beetle-grey, burnt, cut, healthy) would let it map damage pixel by pixel. Test any version by holding out one moor at a time.
6. **PlanetScope** (3 m, near-daily), free under Planet's Education and Research programme, is the intermediate option between Sentinel-2 and aerial imagery.
7. **Replace the fixed threshold with a supervised classifier** once ground-truth data exist.

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A Formulae

Notation: $\rho_k(x, t)$ is the surface reflectance of Sentinel-2 band k at pixel x on date t (digital numbers divided by 10,000). $\mathbf{1}[\cdot]$ is 1 when the condition holds and 0 otherwise.

A.1 Indices

$$\begin{aligned} \text{NBR} &= \frac{\rho_{B8} - \rho_{B12}}{\rho_{B8} + \rho_{B12}}, & \text{NDVI} &= \frac{\rho_{B8} - \rho_{B4}}{\rho_{B8} + \rho_{B4}}, & \text{NDMI} &= \frac{\rho_{B8} - \rho_{B11}}{\rho_{B8} + \rho_{B11}} \\ \text{Brightness} &= \frac{1}{3}(\rho_{B2} + \rho_{B3} + \rho_{B4}), & S &= \frac{\max(\rho_{B2}, \rho_{B3}, \rho_{B4}) - \min(\rho_{B2}, \rho_{B3}, \rho_{B4})}{\max(\rho_{B2}, \rho_{B3}, \rho_{B4})} \end{aligned}$$

The colour saturation S is 0 for a perfectly grey surface and rises with colour.

A.2 Composites

A scene is clear at x if its s2cloudless cloud probability is below 40% and its scene classification is not cloud shadow, cloud (medium or high), cirrus or snow (classes 3, 8, 9, 10, 11). For any index I and window W :

$$I_W(x) = \text{median}_{t \in W, \text{ clear at } x} I(x, t)$$

A.3 Change and seasonal correction

For season s , with the before window W_b (1 July to 30 September) and the after window W_a (16 April to 30 June):

$$\begin{aligned} \text{dNBR}_s(x) &= \text{NBR}_{W_b}(x) - \text{NBR}_{W_a}(x) \\ c_s &= \text{median}_{x \in M} \text{dNBR}_s(x), & \text{dNBR}_s^*(x) &= \text{dNBR}_s(x) - c_s \end{aligned}$$

where M is the moorland in all count areas, sampled at 100 m. The threshold check reports $p_{95,s}$, the 95th percentile of dNBR_s^* over M .

A.4 Detection

$$\begin{aligned} r_s(x) &= \mathbf{1}[\text{dNBR}_s^*(x) > \tau] \cdot \mathbf{1}[\text{NDVI}_{W_b}(x) > 0.3] \cdot \mathbf{1}[x \in M], & \tau &= 0.17 \\ m_s(x) &= r_s(x) \cdot \mathbf{1}[|P_s(x)| \geq 10] \end{aligned}$$

where $P_s(x)$ is the patch of pixels with $r_s = 1$ connected to x through any of its eight neighbours. Ten 10 m pixels are 0.1 ha.

A.5 Areas and percentages

For count area a with area A_a , and pixel area $\omega(x)$ weighted by the fraction of the pixel inside a :

$$H_{a,s} = \sum_{x \in a} m_s(x) \omega(x), \quad \%_{a,s} = 100 \frac{H_{a,s}}{A_a}$$

For a group G of count areas (a moor, a region, or all) over seasons $s = 1, \dots, 9$, the mean annual percentage is area-weighted:

$$\%_G = 100 \frac{\sum_{a \in G} \sum_s H_{a,s}}{\sum_{a \in G} \sum_s A_a}$$

A.6 Beetle damage: rule-based estimate

$$\Delta \text{NBR}(x) = \text{NBR}_{\text{Jul to Sep 2025}}(x) - \text{NBR}_{\text{Jul to Sep 2024}}(x)$$

$$\Delta S(x) = S_{\text{Jul to Sep 2025}}(x) - S_{\text{16 Apr to 30 Jun 2025}}(x)$$

U is the usable ground: moorland, excluding pixels within 20 m of ground detected as burnt or cut in 2024/25 or 2025/26. Each change is centred on its median over U in all count areas (sampled at 20 m):

$$\Delta \text{NBR}^*(x) = \Delta \text{NBR}(x) - \text{median}_U \Delta \text{NBR}, \quad \Delta S^*(x) = \Delta S(x) - \text{median}_U \Delta S$$

$$f(x) = \mathbf{1}[\Delta \text{NBR}^*(x) < -0.10 \text{ or } \Delta S^*(x) < -0.05], \quad R_a = \frac{1}{|a \cap U|} \sum_{x \in a \cap U} f(x)$$

A.7 Beetle damage: gradient boosting

Labels from the survey. With d_a the surveyed damage of count area a , each sampled pixel x in a gets

$$y_x = \begin{cases} 0 & d_a = 0 \\ 1 & d_a \geq 0.5 \\ \text{no label} & \text{otherwise.} \end{cases}$$

Measures. z_x holds 45 changes: 15 measures (bands B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12; NBR, NDVI, NDMI, S , brightness) for each of three comparisons (summer 2025 minus summer 2024; late minus early summer 2025; summer 2026 minus summer 2024).

Model. Gradient boosting (Friedman, 2001), as implemented in scikit-learn (Pedregosa et al., 2011), builds an additive score from M small regression trees h_m (depth 3):

$$F_M(z) = F_0 + \nu \sum_{m=1}^M h_m(z), \quad p(z) = \frac{1}{1 + e^{-F_M(z)}}$$

Each tree is fitted, on a random 80% of the labelled pixels, to the negative gradient of the log-loss

$$L = - \sum_x [y_x \log p(z_x) + (1 - y_x) \log(1 - p(z_x))]$$

with learning rate $\nu = 0.05$ and $M = 150$ trees.

From pixels to a count area. The mean probability $\bar{p}_a = \frac{1}{n_a} \sum_{x \in a} p(z_x)$ is mapped to a proportion with a straight line fitted by least squares on the training count areas, using out-of-fold probabilities:

$$\hat{d}_a = \min(1, \max(0, \alpha + \beta \bar{p}_a))$$

Combined estimate. $\hat{c}_a = \frac{1}{2} (R_a + \hat{d}_a)$.

A.8 Evaluation

Leave-one-moor-out. For each moor m , the model and its calibration line are fitted using only the other moors' pixels and survey scores; $\hat{c}_a$ is then computed for the count areas of m . The moor-wide medians above use imagery from all count areas but no survey scores.

$$\text{Typical difference} = \frac{1}{n} \sum_{a=1}^n |\hat{c}_a - d_a|, \quad \text{Within } k = \# \{a : |\hat{c}_a - d_a| \leq k\}$$

Rank agreement is Spearman's ρ : the Pearson correlation of the ranks of $\hat{c}_a$ and d_a .

Paired bootstrap. For $b = 1, \dots, 5000$, draw n count areas with replacement and compute $\Delta_b = \text{TD}_b^{\text{model}} - \text{TD}_b^{\text{rule}}$, the difference in typical difference on that sample; the 90% interval is the 5th to 95th percentile of Δ_b .

Region-only benchmark. $\hat{d}_a$ is the mean of d over the other moors in the same GWCT region.

Nested model choice. For each held-out moor m , the training moors are split into five groups. Each candidate model's combined estimate is scored by holding out each group in turn; the model with the lowest typical difference is refitted on all training moors, and it alone predicts m .

Region hold-out. As leave-one-moor-out, but every moor in one GWCT region is held out together (North York Moors; southern Dales). The northern Dales region cannot be held out: it contains every count area the survey scored 0.

B Threshold sensitivity

Table 12. Main results at four detection thresholds.

Threshold	Total vs 0.17	Season order ^a	Moor order ^a	NYM / NP	After / before 2021	Moor House %	Geltsdale %
0.13	+61%	0.97	0.98	1.64	1.03	0.65	4.5
0.15	+25%	0.98	0.99	1.72	1.00	0.23	2.0
0.17	main run	1.00	1.00	1.75	0.98	0.08	1.0
0.19	−20%	0.93	1.00	1.78	0.97	0.02	0.6

^a Rank correlation with the 0.17 results. NYM / NP: North York Moors against North Pennines, mean annual percentage. After / before 2021: mean of 2021/22 to 2023/24 against 2017/18 to 2020/21.

Lowering the threshold barely changes the after/before 2021 ratio (0.97 to 1.03), so there is no sign that faint cut scars, which a lower threshold would pick up, cluster after the regulation. At 0.13 the reserves register clearly (Moor House 0.65%, Geltsdale 4.5% a year), which marks 0.13 as too permissive. At 0.19 Geltsdale falls to 0.55%.

C Further results on heather beetle damage

C.1 Earlier attempts

- **Browning alone** (the share of each count area whose NBR fell from summer 2024 to summer 2025): typical difference 0.27, rank agreement 0.60. The whole area browned in the dry summer of 2025, so the moor-wide change has to be removed.
- **Rule-based estimate without the 20 m margin** around burnt and cut ground: typical difference 0.13. Edges of fresh burns were being counted as damage.
- **A heather mask from seasonal greenness** (heather stays green in winter): typical difference 0.15, worse than without it. WorldCover has no heather class in these count areas, so no better mask was available.
- **Calibrating the rule-based estimate** with a straight line fitted on the other moors: typical difference 0.125, no better than the uncalibrated share (0.123), so the rule is used as it stands.
- **A random forest on its own**: typical difference 0.13 as a classifier of healthy and damaged pixels, and 0.16 predicting each pixel's survey score directly. Averaged with the rule it reaches 0.11 (Appendix C.2).

C.2 Six models compared

Table 13. Six models for Step 2, alone and averaged with the rule-based estimate. 22 count areas, leave-one-moor-out. The interval is the paired-bootstrap 90% interval for the change in typical difference against the rule alone; negative is better.

Model	Model alone			Averaged with the rule				
	Typical diff.	Within 0.1	Rank ρ	Typical diff.	Within 0.1	Within 0.2	Rank ρ	90% interval vs rule
Rule-based estimate only	0.122	12	0.81					
Gradient boosting (classic)	0.118	10	0.79	0.098	15	19	0.88	−0.053 to +0.003
Gradient boosting (LightGBM style)	0.122	10	0.78	0.101	15	20	0.87	−0.049 to +0.006
Support vector machine	0.124	11	0.78	0.102	15	20	0.86	−0.049 to +0.008
Random forest	0.141	12	0.78	0.109	14	19	0.86	−0.043 to +0.013
Logistic regression	0.151	9	0.72	0.109	13	19	0.84	−0.043 to +0.018
Extra trees	0.161	9	0.71	0.118	11	19	0.83	−0.032 to +0.022

All six models improve on the rule when averaged with it, with typical differences of 0.10 to 0.12 (Table 13). Classic gradient boosting is best (0.098) and is used in the report. Every 90% interval includes zero: with 22 count areas the gains are consistent across models but individually within chance. More and better labels will matter more than the choice of algorithm. The models were fitted with scikit-learn (Pedregosa et al., 2011): gradient boosting with 150 trees of depth 3, learning rate 0.05 and 80% subsampling, settings fixed in advance.

C.3 Checks for data leakage

Leakage means information about the answer reaching the model, which makes results look better than they will be on new data. The burning and cutting analysis fits nothing to outcomes, so this applies only to the beetle model.

Table 14. Possible sources of leakage and how they are handled.

Possible leak	How it is handled
A moor's own pixels or survey score used to predict it	Never: every estimate comes from a model trained without that moor
Calibration fitted on the model's own training fit	The line uses out-of-fold probabilities, split by moor
Measures built from survey scores	All measures are image changes
Settings tuned on the test results	Model settings were fixed in advance
Moor-wide reference includes the held-out moor	Imagery only, no survey scores; one moor barely moves a median over 4,000 ha
Choosing the best of six models on the same data	Nested test: 0.105 against 0.098; classic gradient boosting chosen in 9 of 15 folds
Similar neighbouring moors in training	Region hold-out (Table 15)

Table 15. Region hold-out: typical difference on the 8 count areas of the North York Moors and the southern Dales.

Estimate	Alone	Averaged with the rule
Rule-based estimate	0.075	
Gradient boosting, one moor held out	0.124	0.059
Gradient boosting, whole region held out	0.215	0.072

Gradient boosting alone relies partly on having seen similar moors nearby; averaged with the rule it holds up when a whole region is unseen (Table 15). The flagging cut-offs and model settings were fixed in advance, but greying as a measure and the decision to average were settled while looking at the same 22 count areas. A second survey year is the clean test.

C.4 Estimates for every count area

Table 16. Surveyed damage and satellite estimates by count area, each moor predicted without its own data. Grinton, Long Gill (survey 0.5) had too few clear images and is not estimated.

Count area	Survey	Rule	Gradient boosting	Combined	Difference
<i>Northern Dales</i>					
Bollihope, Pikestone Fell	0.0	0.14	0.13	0.13	+0.13
Crossgill, Dowks	0.0	0.02	0.17	0.09	+0.09
Crossgill, Pants	0.0	0.19	0.14	0.16	+0.16
Edmundbyers, Waskerley	0.0	0.23	0.17	0.20	+0.20
Geltsdale, Middle	0.0	0.13	0.00	0.07	+0.07
Moor House, Trout Beck	0.0	0.55	0.05	0.30	+0.30
Moor House, Behind House	0.0	0.10	0.01	0.05	+0.05
Stags Fell, Beacon	0.0	0.08	0.00	0.04	+0.04
Crossgill, Greencastles	0.1	0.03	0.24	0.14	+0.04
Wemmergill, Close House	0.1	0.20	0.01	0.11	+0.01
Bollihope, Catterick	0.3	0.41	0.47	0.44	+0.14
Eggleston, Ever Rigg	0.3	0.31	0.29	0.30	+0.00
Eggleston, Blackton	0.3	0.43	0.34	0.38	+0.08
Wemmergill, South Side	0.6	0.37	0.11	0.24	−0.36
<i>North York Moors</i>					
Danby, Low Moor	0.5	0.51	0.61	0.56	+0.06
Westerdale, Tank	0.5	0.54	0.65	0.59	+0.09
Rosedale, Pits	0.6	0.69	0.66	0.67	+0.07
Danby, Glaisdale	0.7	0.76	0.65	0.70	+0.00
Snilesworth, Lodge	0.8	0.92	0.57	0.74	−0.06
<i>Southern Dales and Peak</i>					
Ramsgill, Raygill	0.2	0.22	0.39	0.30	+0.10
Dallowgill, Bishops	0.6	0.78	0.53	0.66	+0.06
Swinton, Potts	0.7	0.78	0.56	0.67	−0.03

Where the survey found little or no damage (0 to 0.1), the combined estimate runs about 0.11 high on average. Survey scores are estimates in tenths, so agreement much closer than 0.1 cannot be

expected.

Heather beetle: the 5 count areas where satellite and survey agree best, for contrast

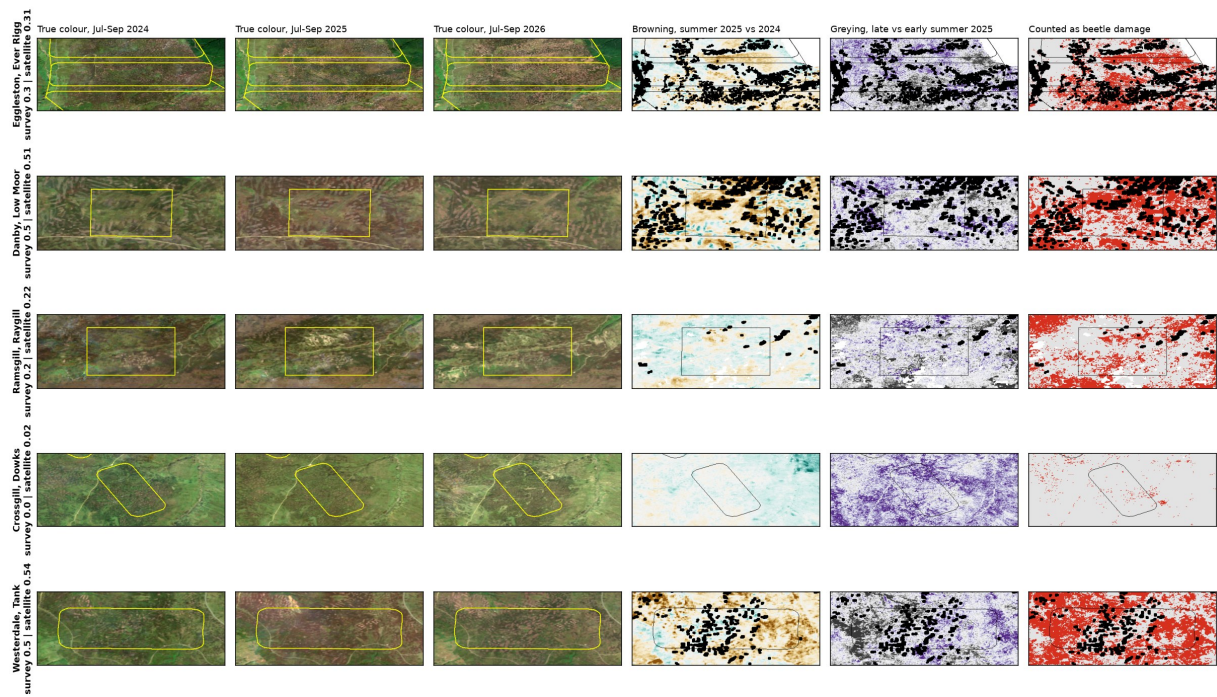


Figure 12. The five count areas where the rule-based estimate and the survey agree best, for comparison with Figure 11.

D Eggleston boundaries

Table 17. Eggleston and the North Pennines as count areas, as the strip between them, and as Eleanor’s single block.

Scope	Boundary	Area (ha)	Mean % a year
Eggleston	count areas	1,557	5.2
Eggleston	strip between counts	371	5.8
Eggleston	count areas + strip (Eleanor’s block)	1,928	5.3
North Pennines	count areas	3,237	3.7
North Pennines	strip between counts	371	5.8
North Pennines	count areas + strip (Eleanor’s block)	3,608	3.9

The strip between Eggleston’s count areas is managed slightly more than the count areas themselves (5.8% against 5.2% a year), so including it would raise the North Pennines figure from 3.7% to 3.9%. None of the findings depends on the choice.

E Further imagery

Snilesworth, Lodge (73 ha): managed ground by season

Sentinel-2 false colour (SWIR, NIR, red). Red = detected burning or cutting. Yellow = count area.

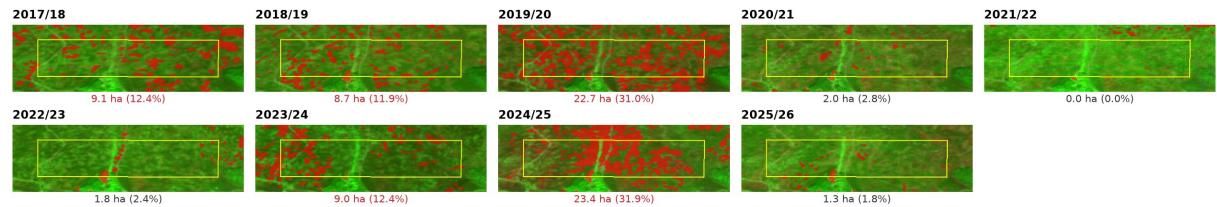


Figure 13. Snilesworth, Lodge: the same sporadic pattern as Waskerley.

Moor House, Behind House (99 ha): managed ground by season

Sentinel-2 false colour (SWIR, NIR, red). Red = detected burning or cutting. Yellow = count area.

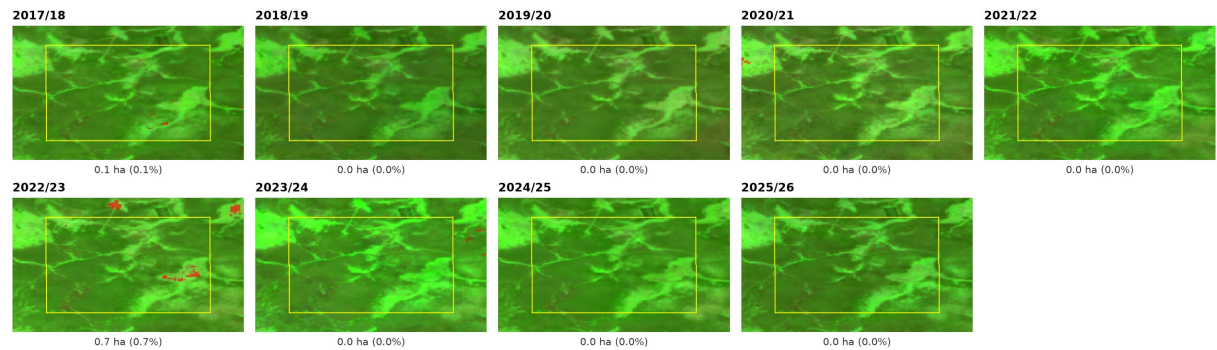


Figure 14. Moor House, Behind House (National Nature Reserve): almost nothing detected in any season.

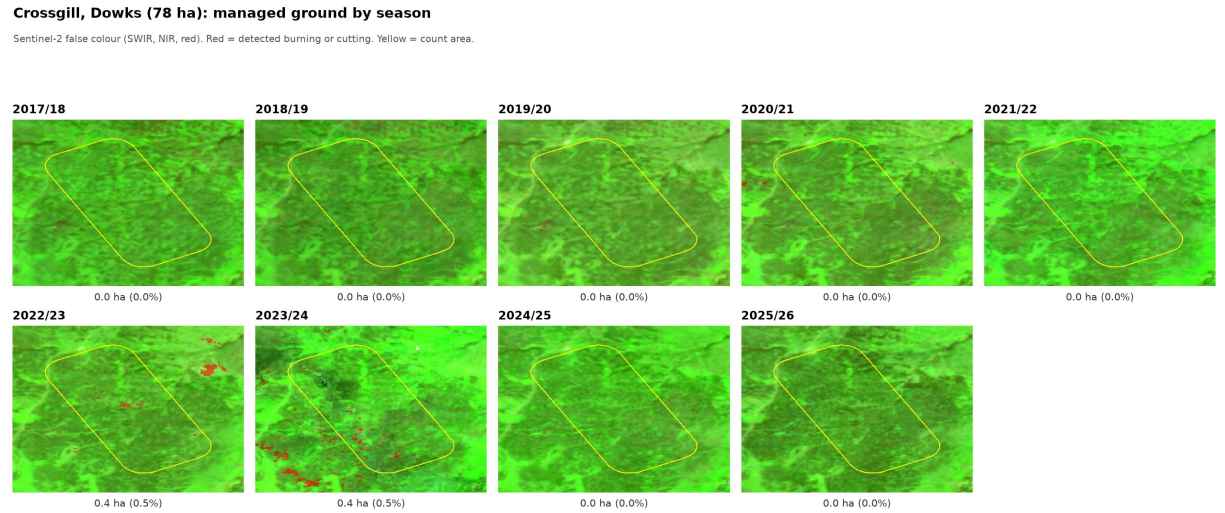


Figure 15. Crossgill, Dowks: at most 0.4 ha in any season. The small cuts PW describes are mostly below the 0.1 ha floor.

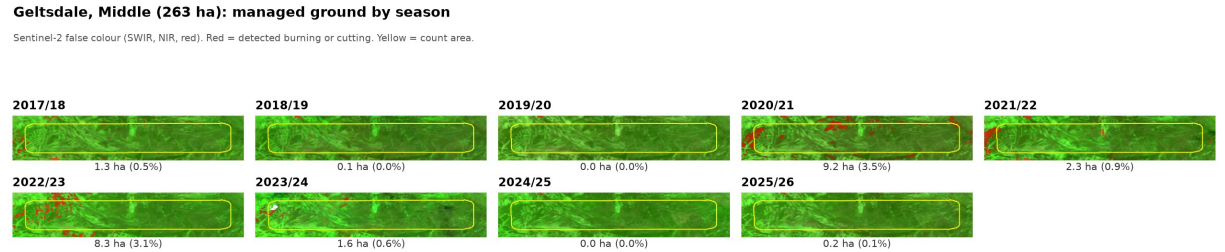


Figure 16. Geltsdale, Middle (RSPB reserve).

F Data and code

Table 18. Files behind this report.

File	Contents
burn_pipeline.py	Detection pipeline (Earth Engine Python API)
build_count_areas.py	Names Eleanor’s polygons from the transect file; splits Eggleston
v5_to_v6_changes.py	Rebuilds version 1’s boundaries; the one-change-at-a-time comparison
beetle_check.py, beetle_rf.py, beetle_model_compare.py, beetle_evidence.py	Beetle damage: rule-based estimate, pixel sampling, gradient boosting and model comparison with leakage tests, image sheets
report_numbers.py, report/make_figures.py	Every number, figure and table in this report, from the outputs
data/count_areas_v6.gpkg	28 count areas plus the Eggleston strip
outputs/results_by_season.csv	Results: 28 count areas by 9 seasons, threshold 0.17
outputs_t013/, outputs_t015/, outputs_t019/, outputs_no_s2c/	Sensitivity runs
outputs_abl_*/, outputs/v5_to_v6/	One-change-at-a-time runs and their summary
outputs/beetle_check/	Beetle damage comparison

File and folder names use the internal numbering, in which this report is v6 and version 1 is v5.